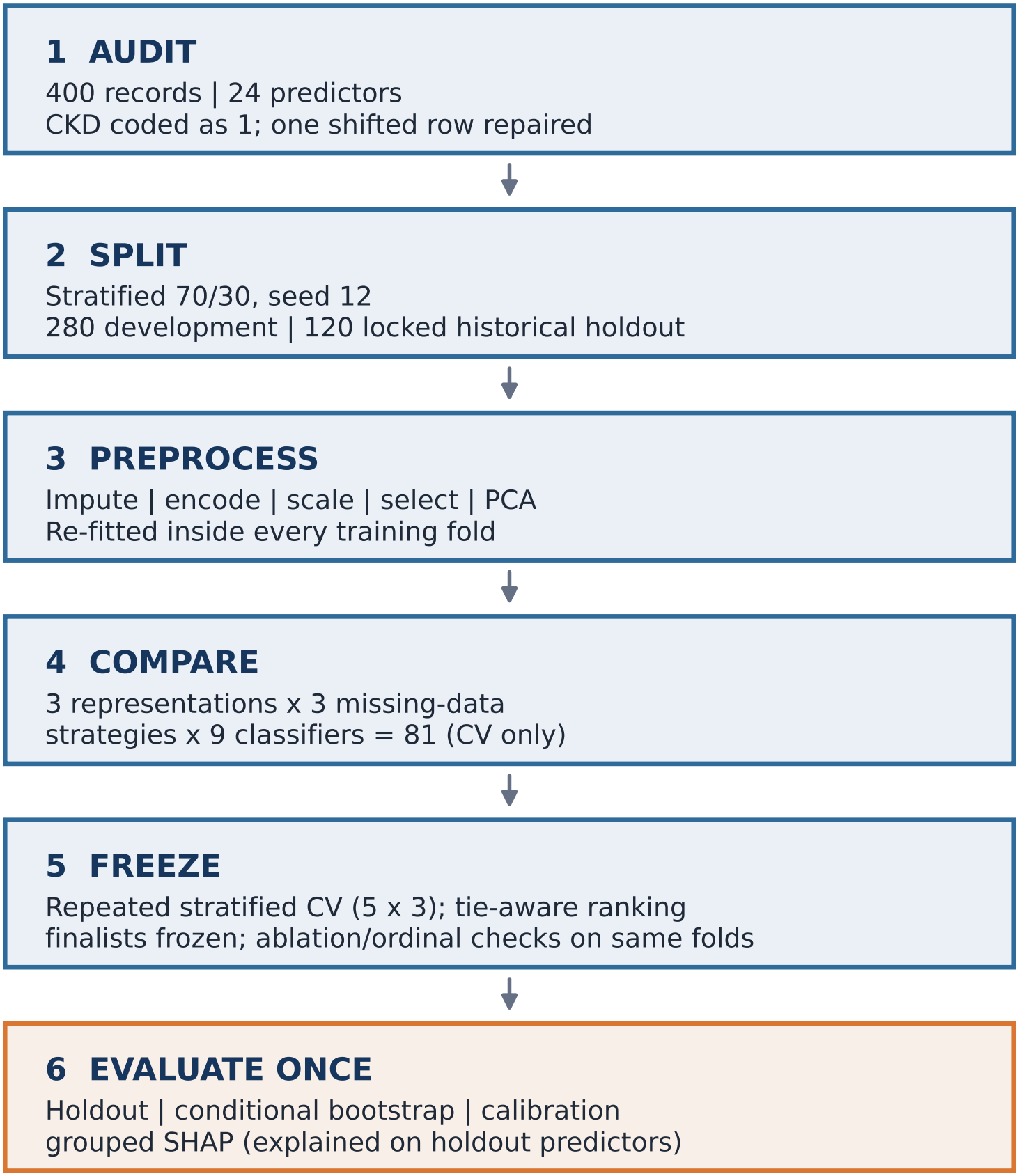
1 AUDIT
400 records | 24 predictors
CKD coded as 1; one shifted row repaired
2 SPLIT
Stratified 70/30, seed 12
280 development | 120 locked historical holdout
3 PREPROCESS
Impute | encode | scale | select | PCA
Re-fitted inside every training fold
4 COMPARE
3 representations x 3 missing-data
strategies x 9 classifiers = 81 (CV only)
5 FREEZE
Repeated stratified CV (5 x 3); tie-aware ranking
finalists frozen; ablation/ordinal checks on same folds
6 EVALUATE ONCE
Holdout | conditional bootstrap | calibration
grouped SHAP (explained on holdout predictors)